

JAYITA SAIKIA

B.Tech (AI & ML) | Patent-Holding Inventor | Product Innovator

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PROFILE

Inventive B.Tech (AI & ML) student at VIT Bhopal University with a **granted design patent**, a filed invention, and 4+ active innovations. Experienced across the full product lifecycle—from ideation and rapid prototyping to IP filing and deployment. Expert at blending software intelligence with hardware design to solve consumer and industrial challenges.

KEY ACHIEVEMENTS

- Granted Design Patent:** Awarded by the Indian Patent Office in June 2025 as a 3rd-year student (App No: 449246-001).
- Filed Invention Patent:** Developed a "Drone with Octo-Arm Mechanism" currently under examination (2026).
- Active Innovation Portfolio:** Managing 4+ simultaneous projects across consumer utility, defense, and assistive technology.

INTELLECTUAL PROPERTY & INNOVATION

Multipurpose Cutting Board | Granted Design Patent

- Designed a multi-functional kitchen tool integrating hygiene and space efficiency for modern consumer environments.
- Executed full patent prosecution, including prior art search, claims drafting, and official journal publication.
- Targeted for scalability within the FMCG and food technology industries.

Drone with Octo-Arm Mechanism | Filed Patent

- Developed an 8-arm robotic gripping mechanism for adaptive payload handling in logistics and aerial robotics.

Origami-Inspired Children's Suitcase | Ongoing Innovation

- Created a foldable, flat-pack travel suitcase designed for safety and accessibility for differently-abled users.
- Engineered a structure that significantly reduces material use and shipping volume for sustainable logistics.

Multi-Bag Holder • *Consumer Utility Design* | Ongoing Innovation

- Ergonomic multi-bag carrying device reducing hand strain; practical everyday consumer utility product.

Sonic-Wave Defence Helmet • *Defence Technology / Wearable Innovation* | Ongoing Innovation

- Sonic wave device integrated into ballistic helmet for real-time threat detection and enhanced combat sensory protection.

Echo-Frequency Navigation stick • *Assistive Technology / Social Innovation* | Ongoing Innovation

- Ultrasonic echo-frequency stick enabling obstacle detection and audio-guided navigation for visually impaired users.

TECHNICAL PROJECTS

Plant Disease Detection System | AI & Computer Vision

- Trained a YOLOv6 model on custom-labeled leaf datasets for real-time disease classification.
- Developed an end-to-end inference pipeline applicable to smart supply chains and quality detection.

Tri-Domain Radar System (Rakshak-3D) | Signal Processing | Ongoing Innovation

- Integrating multi-sensor data fusion to detect threats across air, water, and land domains.
- Designing real-time threat response architecture and antenna signal processing.

SKILLS & CERTIFICATIONS

- **Programming:** Python (Advanced), Java (Proficient), C++.
- **AI/ML:** PyTorch, Computer Vision, YOLOv6, Robo-flow.
- **Design & Prototyping:** SolidWorks, AutoCAD, Sketch-Up, Blender, Rhino.
- **Certifications:**
 - a) **Cloud Computing** — NPTEL — IIT Kharagpur • *Completed*
 - b) **Internet of Things (IoT)** — NPTEL • *Ongoing — Expected 2026*
 - c) **Artificial Intelligence & Machine Learning** — Vidyarthi Platform • *Completed*
 - d) **Networking Essentials** — Cisco Packet Tracer • *Completed*

EDUCATION

B.Tech in Computer Science (AI & ML) | VIT Bhopal University

- 2023 – 2027 (3rd Year)

LANGUAGES

- English (Fluent), Hindi (Proficient), Assamese (Native)
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